RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 17 September 2026, 8:12 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:13 AM
Time taken	1 min 35 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int num;
    int count = 0;

    // Input: A positive integer num
    if (scanf("%d", &num) != 1) {
        return 0;
    }

    count++; // 1 for the loop initialization (i = 1)
    for (int i = 1; i <= num; ++i) {
        count++; // 1 for the condition check (i <= num)

        if (num % i == 0) {
            count++; // 1 for the printf execution inside the if statement
        }

        count++; // 1 for the loop increment (++i)
    }

    // Output: Print the value of the counter variable
    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔

	Input	Expected	Got	
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course